



DEPARTMENT OF THE AIR FORCE
AIR FORCE LIFE CYCLE MANAGEMENT CENTER (AFLCMC)
CRYPTOLOGIC AND CYBER SYSTEMS DIVISION
JOINT BASE SAN ANTONIO-LACKLAND, TEXAS 78243-7081

REQUEST FOR INFORMATION
UNIFIED PLATFORM DATA VALUE STREAM
2 October 2024

- 1) This Request for Information (RFI) is issued in accordance with the Federal Acquisition Regulation (FAR) clause 52.215-3 (Request for Information or Solicitation for Planning Purposes - Oct 1997) and is published to obtain information for use by the Air Force Lifecycle Management Center (AFLCMC), Cryptologic and Cyber Systems Division (CCSD), Joint Base San Antonio - Lackland, Texas, for market research and planning purposes only in support of the Development, Security and Operations (DevSecOps) of the Data Value Stream (DVS) for Unified Platform (UP). UP is responsible for developing and sustaining a cloud-based platform that enables cyber operations across the Department of Defense (DoD).
- 2) THIS RFI IS NOT A REQUEST FOR PROPOSAL, INVITATION FOR BID, OR AN ANNOUNCEMENT OF A SOLICITATION; IT IS ONLY INTENDED FOR INFORMATION OR PLANNING PURPOSES. There is no bid package or solicitation document associated with this announcement. Response to this RFI is strictly voluntary and will not affect any potential offeror's ability to submit an offer if a solicitation is released. Any requests for a solicitation package will be disregarded. The Government does not intend to award a contract based on this RFI or otherwise pay for the information solicited. No entitlement to payment of direct or indirect costs or charges by the Government will arise as a result of preparing submissions in response to this RFI and the Government's use of such information. Submittals will not be returned to the sender. Respondents to this RFI may be asked to provide additional information/details based on their initial submittals.
- 3) Our organization is currently leveraging Rapid Analytic Deployment and Management Framework (RADMF) for UP Big Data Platform (BDP) orchestration on AWS GovCloud. In an effort to identify potential RADMF alternatives, we are conducting market research to evaluate products and solutions that can provide the capabilities to meet the objectives and requirements listed in paragraph 4. The organizational objectives of this research are as follows:
 - a) Identify and catalog available products and solutions in the market.
 - b) Evaluate the features, functionality, and pricing of each alternative to determine its suitability for our needs.
 - c) Assess the level of integration and compatibility of each alternative with our DevSecOps platform, while ensuring compliance with DoD security requirements.
 - d) Estimate the level of government resources required to migrate to a new solution, while maintaining our current operational capabilities.
- 4) The following are some objectives potential solution must help UP achieve:
 - a. Cloud Provisioning – automated provisioning of a cloud-based multi-node platform capable of deploying clusters of over 15 petabytes in size.
 - b. BDP automated deployment – Rapidly and automated provisioning of Big Data Platform Clusters through Continuous Integration/ Continuous Delivery (CI/CD) orchestrated

pipelines.

- c. Achieve desired levels of elasticity and scalability of cloud-based big data platform to accelerate the ingest normalization, and indexing of both unstructured and structured data.
 - d. Data Aggregation and Processing – Confidently store, analyze, and query structured and unstructured data while ensuring data is secure, private, tracked, accurate, and usable.
 - e. Network Data Management - processing engine to handle network packets (PCAP) for observability and threat hunting
- 5) The following are some of the key characteristics that we are using to meet our organizational objectives, as outlined in paragraph 3. While this list is not exhaustive, it does provide a snapshot of some of the most important requirements for our environment:
- a. The solution shall provide orchestration/automation systems and tools which works for NIPRNet, SIPRNet and JWICS.
 - i. The solution shall direct specific computers or virtual machines (VMs), networks, applications, and storage functions from end to end provisioning activities.
 - ii. The solution shall provide auto-scaling functions to maintain a range of numbers of computers or VMs in services on the templates.
 - iii. The solution shall have a web-based, remote management console and role-based control access.
 - iv. The solution must be integrated with a web-based Enterprise Directory.
 - b. The system shall support end-to-end provisioning capabilities
 - i. The solution shall process infrastructure changes, requests to configuration management databases, and changes to subordinate orchestrators across multiple domains.
 - ii. The solution shall support a specific range of numbers of computers or VMs, networks, applications, and data flows and storage.
 - c. The system shall support automated provisioning of workflows
 - i. The solution shall have a workflow design and customization capability.
 - ii. The solution shall have the ability to override requests and/or stop a workflow as required by mission need.
 - iii. The solution shall provide workflow features to make discovery processes more efficient.
 - iv. The solution shall provide activity logs, including successful and failure activities.
 - v. The solution shall provide acceptable performance, that it doesn't impact or degrade existing system performance.
 - vi. The solution shall work to provide partial system orchestration/automation functions without impacting the existing, active system's services and performance.
 - d. The system shall support resource provisioning, configuration, monitoring, scaling and removal.
 - e. The system shall be deployed in a high availability architecture.
- 6) Our organization will consider both commercial and open-source solutions.

- 7) As indicated above, this is market research. Contractor responses will be used for planning purposes. Address any comments and/or questions regarding this RFI in writing by e-mail directly to the Contracting Officer, Ms. Margaret Howsmon, Email: margaret.howsmon@us.af.mil, the Contract Specialist, Ms. Anissa Davis, Email: anissa.davis@us.af.mil, and the Program Manager, Mr. Daryl Banks, Email: daryl.banks.1@us.af.mil
- 8) **Response due date: Wednesday, 16 Oct 2024, 5:00 PM CST**. Please annotate the subject line of e-mail responses as: “BDP Orchestration for Unified Platform RFI #1 Response-‘*contract vehicle*’”. Please send responses to the email to the addresses listed above. Include your contract vehicle information in the subject line of email response along with your CAGE, and DUNs within the message or attachment.